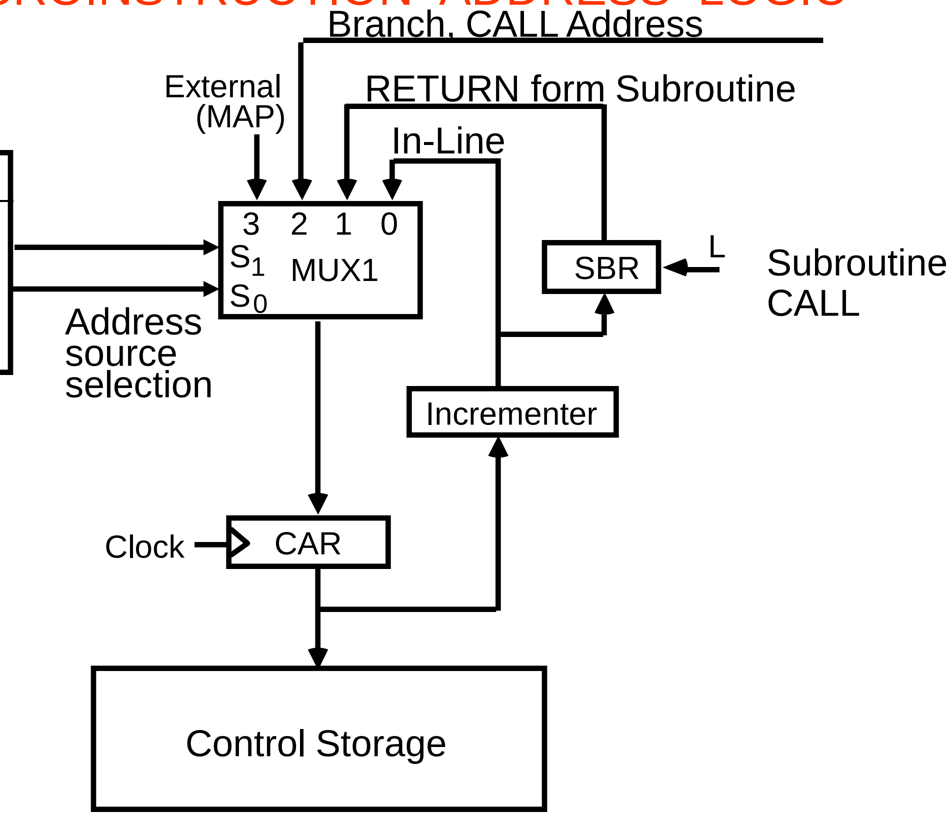
MICROPROGRAM SEQUENCER

- NEXT MICROINSTRUCTION ADDRESS LOGIC -

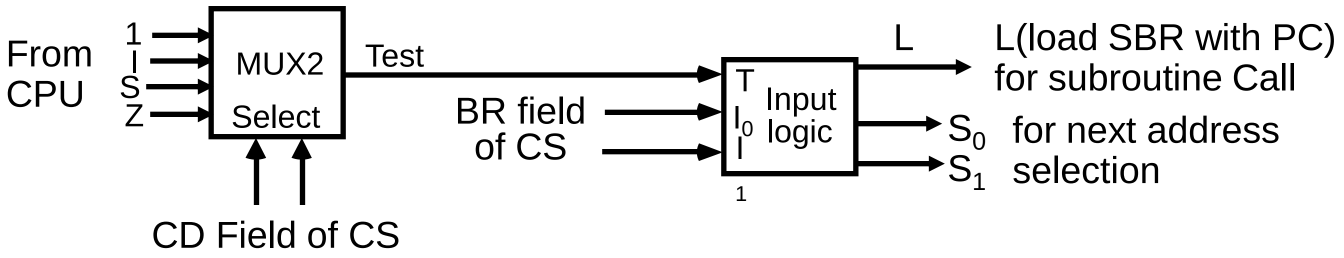
S_1S_0	Address Source
00	CAR + 1, In-Line
01	SBR RETURN
10	CS(AD), Branch or CALL
11	MAP



MUX-1 selects an address from one of four sources and routes it into a CAR

- In-Line Sequencing → CAR + 1
- Branch, Subroutine Call → CS(AD)
- Return from Subroutine → Output of SBR
- New Machine instruction → MAP

MICROPROGRAM SEQUENCER
- CONDITION AND BRANCH CONTROL -



Input Logic

I_1I_0T	Meaning	Source of Address	S_1S_0	L
000	In-Line	CAR+1	00	0
001	JMP	CS(AD)	01	0
010	In-Line	CAR+1	00	0
011	CALL	CS(AD) and SBR <- CAR+1	01	1
10x	RET	SBR	10	0
11x	MAP	DR(11-14)	11	0

$$S_1 = I_1$$
$$S_0 = I_1I_0 + I_1'T$$
$$L = I_1'I_0T$$

MICROPROGRAM SEQUENCER